

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                | Marks available |                 |          |           |          |          |
|----------|-----|------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------|----------|-----------|----------|----------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                | AO1             | AO2             | AO3      | Total     | Maths    | Prac     |
| 6        | (a) |      |  | <b>Any 3 ×(1) from:</b> <ul style="list-style-type: none"> <li>- The particles get closer together / less room to move around / more densely packed</li> <li>- They collide more frequently</li> <li>- Collide [more frequently] with the walls [of the container]</li> <li>- More collisions mean more force [on less area so more pressure]</li> </ul>                       | 3               |                 |          | 3         |          |          |
|          | (b) | (i)  |  | <b>Any of the following calculations earns the 1<sup>st</sup> mark:</b><br>$\frac{20}{-223} = -0.09$<br>$\frac{40}{-173} = -0.23$<br>$\frac{80}{-73} = -1.1$<br><b>A second calculation plus 'do not agree' earns the 2<sup>nd</sup> mark</b><br>DO NOT penalise omission of – sign.<br><b>Award 2 marks for:</b><br>This is only true if the temperatures are in {kelvin / K} |                 |                 | 2        | 2         | 2        | 2        |
|          |     | (ii) |  | Extend line back (1)<br>Read the $T$ when $\{V = 0 / \text{where line crosses the } T \text{ axis}\}$ (1)                                                                                                                                                                                                                                                                      | 2               |                 |          | 2         |          | 2        |
|          | (c) |      |  | $T_2 = 67 + 273 = 340$ (1)<br>Substitution: $\frac{2800}{280} = \frac{V_2}{340}$ <b>ecf</b> (1)<br>Rearrangement: $V_2 = 10 \times 340$ (1)<br>$= 3400 \text{ [cm}^3\text{]} (1)$<br>Award 3 marks for an answer of 26 800 [cm <sup>3</sup> ] from use of 67°C                                                                                                                 | 1               | 1<br><br>1<br>1 |          | 4         | 4        |          |
|          |     |      |  | <b>Question 6 total</b>                                                                                                                                                                                                                                                                                                                                                        | <b>6</b>        | <b>3</b>        | <b>2</b> | <b>11</b> | <b>6</b> | <b>4</b> |